

RESEARCH NOTES

Outline of topics

- Research topic
- Statement of the problem
- Research objectives
- Hypothesis
- Literature review
- Data
- Population and sample

Detailed Content

Research Topic (Title); this refers to the title of the study, which will provide guidelines to the researcher on the scope of the research.

Statement of the Problem; Statement of the problem in research is an issue or concern that puzzles a researcher and hence needs a solution. This should be clearly explained what is to be investigated and how big the problem is.

For example: A researcher may want to find out why enrolment of learners with disabilities is still low in UPE schools despite provision of funds by the government to make the learning environment disability friendly. Statement of the problem should be accompanied by explanation from the background of the study in order to understand clearly the factors that make it to be a problem and justification for studying it.

From the statement of the problem you can develop research objectives, you should also include why the problem is important and hence the need to be addressed.

For example: how enrolment of girls in special education is a problem because it reduces the chances of achievement of gender balance in education.

Statement of the problem explains a definite issue that needs a solution because the problem stands out clearly and is easily recognized.

Research Objectives; A research objective is a specific and measurable statement or relating to the defined purpose of the study. It is therefore a means by which the purpose of the study could be achieved.

Objective in research is important because it outlines specific activities that the researcher wants to achieve by undertaking the study.

For example: The purpose of the study may be “To determine the factors contributing to the low enrolment of learners with disabilities in universal primary schools in Kampala”. From this, one of the objectives that can be formulated is “To determine the number of learners with disabilities in UPE schools in Kampala”.

Secondly, if the problem identified is “Poor Attitudes of Kampala Residents towards Permanent Charity Gaming Clubs”. The general objective of the study could be “To identify the reasons for poor attitudes of Kampala residents towards permanent charity. (A study into the determination of the attitude of Kampala residents towards permanent charity gaming clubs”. Then this can be lit up in the following specific

objectives.

1. To identify the full range of economic, psychological, health and other needs of Kampala guidelines residents on permanent charity gaming.
2. To determine how improved performance of permanent gaming clubs can contribute to concern the attitudes of residents of Kampala.



3. To determine the extent to which formal and informal support systems address these needs from the view point of service providers as well as residents.

Thus the purpose is a general statement of the aim of study while the objectives must be SMART. That is they must be;

S - Specific

M- measurable

A- achievable

R - realistic

T- time bound.

Importance of Research Objectives

The formulation of objectives helps in the following areas;

1. Focus the study (narrowing it down to essentials)
2. Avoid the collection of data which are not strictly necessary for understanding and solving the problem you have identified.
3. Properly formulated, specific objectives will facilitate development of your research methodology and will help to orient the collection, analysis, interpretation and utilization of data.

Research Questions; Research questions are issues that a researcher intends to answer so as to achieve the study objectives. Research questions should be related to research objectives. It is the objectives that the researcher translates into research questions.

For example: “To determine the number of learners with disabilities in UPE schools of Kampala District”. Can be translated into a research question such as; “What is the number of learners with disabilities in UPE schools of Kampala District?”

Hypothesis Development; A hypothesis is a suggested explanation of a

phenomenon, or alternatively a reasoned proposal suggesting a possible correlation between or among a set of phenomena.



A hypothesis is formulated in such a way, that this hunch (an idea that based on a stronger feeling that one has and not evidence) can be tested. Based upon the findings of the research, the hypothesis is supported or rejected and more hypotheses are formulated to continue the process of building cohesive theory. The most common use of hypothesis is to test whether an existing theory can be used to solve a problem. In everyday situations, those who confront problems often propose informal hypotheses that can be tested directly.

For example, when a bulb fails to light when the switch is turned on, several hypotheses come to mind based upon our understanding of electricity and our past experiences with bulbs.

1. The plug is not properly connected to the wall outlet.
2. The bulb is buried out
3. The fuse is burned out or the circuit breaker has been tripped.
4. There has been a power failure in the neighborhood.

Each of these speculations can be tested directly by checking the plug connection, substituting a bulb known to be in working condition, inspecting the fuse or circuit breaker, or by noting whether or not other lights in the house or in neighbor's house are on. Therefore, a research or scientific hypothesis is a formal affirmative statement predicting a single outcome.

Literature Review; This is the information got from publications that is related to the problem under investigation. A literature review is much more than a collection of different extracts. It must be well organized preferably according to the different themes, sections or variables of the study questions or the objectives of the study.

Data; In research, data refers to information collected or gathered by a research from the field or existing sources with an aim of finding a solution to the research problem. Data analysis; this involves critical examination of collected data as to generate

required information out of it.



This refers to the stages the researcher passes through if he is to make a sound and conducive research. The different steps guide the researcher right from the problem identification or definition to the communication of results. A research without direction is like a river without a source. Any research carried out follows a systematic process of eleven steps.

1. Identifying a problem and defining it
2. Stating the objective of the study
3. Developing of hypothesis
4. Reviewing the related literature
5. Choosing and designing research methods (research design)
6. Designing questionnaires
7. Carrying out a pilot study and sampling
8. Organizing of resources
9. Conducting the research
10. Processing the results
11. Reporting the results.

The research process includes the systematic identification, collection, analysis and distribution of information for the purpose of knowledge development and decision making. Whether you are creating a new research programme or perhaps revising an existing research programme, what are the steps you should take? While there are dozens of little steps along the way, each of those steps fits into one of the 6 major steps of the marketing research process; they are

Step 1: Identifying and Defining your Problem

Step 2: Developing your Approach

Step 3: Establishing Research Design and Strategy

4: Collecting the Data

Step 5: Performing Data Analysis

Step 6: Reporting and Presentation.



Step 1: Identifying and Defining your Problem

This precedes all other activities in the research process. Nothing meaningful can take place not unless there is an issue to investigate. One needs to know first what to research about. It is through this process that one answers the questions of “what?”, “why?” and “How?” the problem should be clearly stated and a hypothesis developed. When selecting a research problem, a researcher should keep in mind the availability of times money, subject of study and equipment needed. Defining a problem is all about clarifying issues and aiding the research to focus attention on what is relevant to the problem. It paves the way to proper course of action, uncovers the nature and boundary of the problem and gives guidance to the researcher to design appropriate methodology and sampling methods.

Techniques Involved in Defining a Problem

A research problem should be defined systematically giving due weight to all relating points.

The following steps should be followed:

1. Statement of the problem in a general way while understanding the native of the problem.
2. Developing the ideas through discussions and rephrasing the research problem
3. Technical words or phrases with special meaning should be clearly defined and basic assumption relating to the research problem clearly stated.
4. A straight forward statement of the value of the investigation i.e the criteria for the selection of the problem should be provided.
5. The suitability of the time-period and the sources of data available must be considered by the researcher in defining the problem.
6. Overdone topics should not be normally chosen.
7. Controversial subjects should be avoided.
8. Narrow and vague problems should be avoided.

9. The selected subject should be familiar and feasible so that the related materials and sources are within one's search.



10. The researcher should consider the practicability of the study concerned i.e he should be in position to positively answer the following questions;

- a) Is he well equipped in terms of his background to carry out the research?
- b) Does the study fall within the study he can afford?
- c) Are the respondents willing to cooperate or participate in the research?

11. Stating the goals and objective of the study.

12. Once the researcher has introduced the general area, narrowed it down to specific areas, then comes the task of stating the purpose of the study, which is the central point of any research.

If you are considering conducting research, since the problem has been already identified, then there is an information need. This step is always the start of the research process. At this point, the point will have been recognized by at least one level of management and internal discussions will have taken place. Sometimes, further definition of the issues and tools is needed in internal and external secondary research.

Leading examples for external secondary research sources include;

- Newspapers / magazine articles (business and trade shows)
- Television
- Newsletters
- Competitive intelligence firms
- Industry reports
- Business directories
- Trade associations
- Government publications and websites
- Search engines (www.google.com)
- Friends and colleagues

Step 2: Developing your Approach

Once your problem is well defined, you can move onto developing your approach will generally be around a defined set of objectives. Clear objectives developed in step I will



lead to better development. Developing approach consists of research skills such as; establishing a budget, understanding your environment and its influencing factors, developing an analysis model and formulating hypothesis.

Step 3: Research Design

Based upon a well-defined approach from step 2, a frame work for the designing research programme should be apparent. This step is the most encompassing of all steps in the research process, requiring the greatest amount of thought, time and expertise and is the point at which those less experienced with research will obtain assistance from a research expert. Since the intelligence eventually gained from the research is so closely related to the selected research design, this is the single most important step in the research process and most vulnerable to the typical errors. This includes elements of research design.

- The questionnaire designing
- Measuring and scaling
- Sample size calculation
- Basic statistical testing

Step 4: Collecting Data

It's often called data collection or survey fielding. This is the point at which the finalized questionnaire (survey instrument) is used in gathering information among the chosen sample area. There are a variety of data collection methodologies to consider.

Selecting which is the most appropriate data collection methodology for a particular research takes place during steps 2 & 3 of the market research process.

Data collection typically begins with field testing of the final questionnaire with a small portion of the respondent sample to make sure it is gathering information

correctly. Regardless of the data collection methodology chosen, the data collection process often takes from 25 percent to 50 percent of the total time needed to complete a research project. Research data collection methods include;

1. Computers assisted telephone interview



2. Internet survey
3. Mail survey
4. Traditional telephone interviewing

Step 5: Performing Data Analysis

Any questionnaire data analysis will depend on how the questionnaire was constructed. Less complex questionnaire data analysis can be handled with any of a number of office suite tools, while more complex questionnaire analysis requires dedicated research analysis programmes. Types of statistical data analysis that might be performed are simple frequency distributions, cross tab analysis, multiple regressions, etc. The more complex the needed level of statistical data analysis is, the more time and cost it will take to execute.

Step 6: Reporting and Presentation

Research reporting a presentation is lastly and the, second most important step. Any business critical information and knowledge that comes from your research will be limited by how it is presented to decision makers. All the information that will help convince the decision makers should be clearly presented to them giving a base on which they can account for the decisions so made.

Literature review is the analysis of textbooks, or manuscripts that a researcher consulted in understanding and investigating the problem during the research. This information about the research problem is needed so that it can be put in the proper context for the research to proceed effectively.

Basically in literature review the researcher attempts to determine what other scholars have found out about similar research problem in order to gather information relevant to the research at hand.

Literature review is a critique of the literature on the chosen variables / constructs cited in the study (relational review). It is not meant to contain all the literature on a given subject, only that relevant to the context of the research and should identify what has been done and what needs to be done. It is a systematic process that requires careful and perceptive reading and attention.

Importance of Literature Review

It enables the researcher to;

1. Know what has been done in a particular areas or selected field.
2. Identify and create limit what needs to be done.
3. Sets ground for further studies
4. Identify the strength and weakness of other researcher.
5. It is a basis for developing historical background.
6. Controversies could easily be identified and reconciled.
7. Avoid duplication of knowledge
8. Know the recommendations of previous researchers for further research.
9. Understand research methodology.
10. Provides background for interpreting results of the study.
11. Researcher builds confidence in the area of concern.

12. It is a basis for developing historical background.



Steps Involved in Literature Search

Steps involved in literature search

These include;

- Identifying the key issues in the research problem to be addressed by the literature review. These may include the research title, research question and objectives.
- Identifying sources of information such as index or retrieval system. For example Libraries and internet.
- Identifying titles of relevant reports, periodic papers and related research publications.
- Separate the related academic material reports into categories of relevance and delete irrelevant reports, analyzed related academic material relevant to the current study which shows current areas in the study.
- Preparing abstracts or summaries of the related academic materials.
- Writing the review of literature by creating the outline of the major points and arranging the literature in headings and subheadings.
- Preparing the list of references from the research reports reviewed.

Sources of Related Literature

The library is the major source of literature, it may house published books, periodicals, technical reports, academic these and internet services.

These are classified into three main sources of literature review.

1) General Sources

These are references that researchers turn to first. They direct researchers to relevant sources.

They include the following;

- a. Indicates that List authors, title of article, publishers, place of publication and year of publication.

- b. Abstract that gives a brief summary of various publication, year of publication as well as their author, title of article and place of publication for example “current index to journals in education” and “psychological abstract”.



2) Secondary Sources

These are publications in which authors describe the works of other people. The author is not a direct observer of participant in the events that are described. The most common sources include; text books, modules, dictionaries and encyclopedias, etc.

3) Primary Sources

These are publications in which investigations report the results of their studies. The researcher communicates findings directly to the reader. Most primary sources are journals, newspapers, bulletins and research reports and circulars.

Review of literature should be based on primary source because sometimes writers of secondary sources alter original information.

Assignment:

In your area of interest, please download any research article and summarize it in half a page for your inclusion in your literature review section of your research. Appropriate citation and referencing is a must.

As discussed at the beginning of this subject, this week will be dedicated to students working on their project.

In your area of interest, you are required to come up with the following areas of a concept paper:-

- i) Research Topic
- ii) Problem Statement
- iii) Research Purpose
- iv) Research Objectives
- v) Research Questions
- vi) Brief review of literature related to your study



READING MATERIALS

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